

Technical Document #917: Data Engineering - Comprehensive Overview

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Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Data warehouses store structured data optimized for analytical queries.
- Schema evolution handles changes in data structure over time.
- Stream processing handles continuous data streams in real-time.